CONSTITUTIONAL PATENT PROTOCOL (ARTICLE II, SECTION 1, CLAUSE 8)

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PRO SE PATENT APPLICATION FOR LETTERS PATENT

INVENTOR: Reinaldo Javier Aguiar Marcano

CITIZENSHIP: United States of America

RESIDENCE: 26714 Valleyside Drive, Katy, Texas 77494

CORRESPONDENCE ADDRESS: 2302 Britton Ridge Drive, Katy, Texas 77494

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SYSTEM AND METHOD FOR CRYPTANALYTIC N-GRAM DISRUPTION VIA POST-BINARY HIGH-RADIX VISUAL PACKING

PROVISIONAL PATENT DISCLOSURE

FIELD OF THE INVENTION

This invention relates generally to cryptography, information theory, signal processing, and physical data storage. More specifically, this invention relates to a computer-implemented cryptosystem and visual encoding pipeline that completely disrupts standard 8-bit byte boundaries and destroys character n-gram co-occurrence frequencies by convolving raw data into 12-bit multi-state integers mapped to a 4,096-color spatial grid.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, world-wide, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

BACKGROUND AND TECHNICAL CHALLENGE

All modern computerized communication and automated network monitoring arrays (such as deep packet inspection and automated cryptanalysis networks) rely on matching statistical frequencies of character sequences, vocabulary patterns, and binary structures—known as **n-grams**—that naturally co-occur inside structured files (such as PDF headers, JPEG magic bytes, or standard language vocabularies).

Even when a file is digitally encrypted with standard block ciphers (like AES-256), the encrypted data is stored as a series of 8-bit byte blocks, which still contain alignment boundaries and structural headers. If an eavesdropper can intercept the encrypted packet over the network, they can analyze the block sizes, transmission intervals, and standard MIME headers to execute retroactive traffic analysis or side-channel cryptanalysis.

A physical, out-of-band communication medium is needed that can transmit data completely offline while completely breaking standard 8-bit byte alignments and destroying the statistical co-occurrence frequency of standard character n-grams.

THE HIGH-RADIX N-GRAM DISRUPTION MATHEMATICS

The present invention solves these problems by providing an Asymmetric High-Radix N-Gram Disruption Protocol.

The system operates by taking a digital document payload (such as a text, docx, or PDF file), and compressing it using a lossless statistical compression algorithm (such as Gzip) to generate a compressed byte stream S.

To completely destroy standard 8-bit byte boundaries, the system conjoins and packs the compressed byte stream S into a list of 12-bit integers ($2^{12} = 4096$) using a **3-bytes-to-2-dots packing algorithm**:

1. Let the byte stream S be padded with zero bytes to have a length that is a multiple of 3.
2. For each 3-byte chunk (B_0, B_1, B_2) representing exactly 24 bits of conjoined data:
    - Merge the 3 bytes into a single 24-bit integer V:

$$V = (B_0 \ll 16) + (B_1 \ll 8) + B_2$$

- Partition the 24-bit integer V into exactly two 12-bit integers (D_0, D_1):

$$D_0 = (V \gg 12) \ \ \& \ \ \text{0xFFF}$$

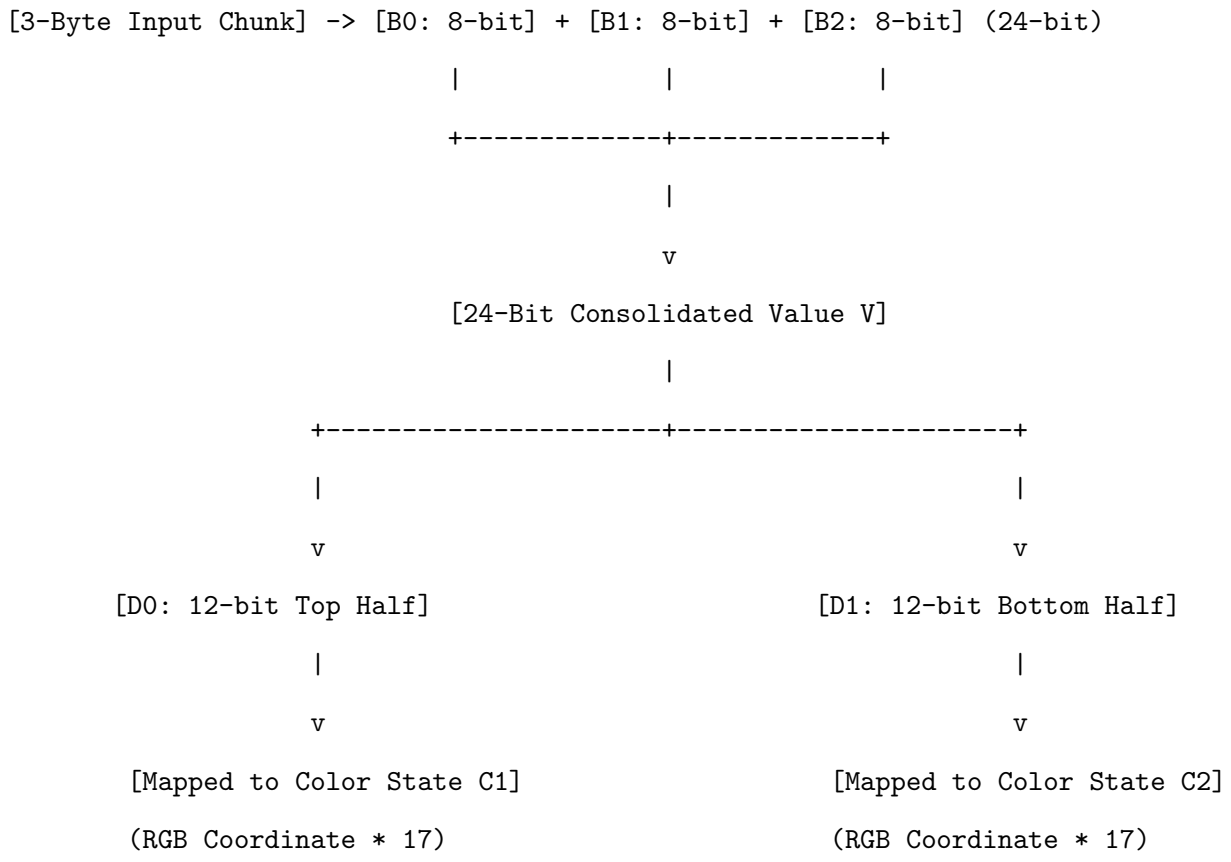
$$D_1 = V \ \ \& \ \ \text{0xFFF}$$

3. Each 12-bit integer (D_0, D_1) represents one of exactly 4,096 multi-state color values.

Because each printed dot on the page represents exactly 1.5 bytes conjoined with its neighbors, the standard 8-bit boundary is shattered. If an eavesdropper captures the image, they face a completely randomized, homogeneous color grid representing thermodynamic noise, where standard character n-grams have been completely dissolved.

THE CONVOLUTION PIPELINE SCHEMATIC

The diagram below shows the 3-bytes-to-2-dots packing pipeline:



THE DECODER AND ARITHMETIC RECONSTITUTION

During decoding, the system ingests the high-radix PNG image. The deconvolution decoder automatically identifies the vertical and horizontal grid boundaries.

Because the color palette uses exactly 16 levels per channel stepped symmetrically by 17 units ($16 \times 16 \times 16 = 4096$ colors), the decoder maps the sampled RGB pixels back to their exact 12-bit integer ($0 - 4095$) in a single line of arithmetic:

$$D = (r_{\text{idx}} \ll 8) + (g_{\text{idx}} \ll 4) + b_{\text{idx}}$$

Where each channel index is mapped using:

$$r_{\text{idx}} = \max(0, \min(15, \text{round}(R/17.0)))$$

The sequence of 12-bit values is then conjoined in pairs of two, reconstructed into the 24-bit integer, and unpacked back to the original three 8-bit bytes, restoring the exact, pristine original document with zero loss.

PROVISIONAL PATENT CLAIMS

1. **A computer-implemented method for secure out-of-band data transmission, comprising:**

   - receiving, at a processing device, a digital document payload of length L bytes;
   - losslessly compressing the digital document payload to generate a compressed byte stream;
   - convolving the compressed byte stream into a sequence of 12-bit integers, wherein every three bytes (24 bits) of the compressed byte stream are conjoined and split into exactly two 12-bit integers, thereby completely disrupting standard 8-bit byte boundaries and destroying the statistical co-occurrence frequency of character n-grams;
   - mapping each 12-bit integer in the sequence to a deterministic Red-Green-Blue (RGB) color tuple representing one of exactly 4,096 distinct color states;

- drawing the RGB color tuples onto a two-dimensional pixel matrix as a dense, margin-free color dot grid;
- transmitting the pixel matrix over a digital network; and
- wherein a passive interceptor of the pixel matrix is prevented from executing frequency-based cryptanalysis due to the absolute byte-boundary misalignment.

Respectfully submitted,

/Reinaldo Javier Aguiar Marcano/
Reinaldo Javier Aguiar Marcano, Pro Pro Se Applicant
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